

Atomic Data in EXHALE versus MoCHII: Comparison and the 2026-07-17 Rate Update

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1. Purpose and scope

This memo has two parts. Part 1 compares the atomic data that EXHALE and MoCHII (a separate code, not part of this repository) use for the microphysical processes the two codes have in common: photoionization cross sections, recombination, collisional ionization, charge exchange, cooling, and heating. The conclusion is that, where the two codes overlap, they draw on the same lineage of published data; the differences are in coverage (MoCHII spans more elements; EXHALE adds the metastable He I 2³S network and planet-atmosphere species) and in a few modeling choices.

Part 2 documents the rate update applied on 2026-07-17. Before that update EXHALE inherited its H/He recombination and collisional-ionization fits from ATES, and several of those fits carried no source annotation. The update makes the EXHALE default match the Badnell/Mao recombination and Voronov collisional ionization that MoCHII uses,

introduces the `legacy_hhe_rates` switch to reproduce the old fits, replaces the free-free Gaunt factor with the van Hoof et al. (2014) thermal average, and corrects a physical error in the free-free charge weighting.

2. Part 1 — Process-by-process comparison

Table 1 summarizes the overlapping processes. The subsections below add the details that do not fit in the table.

2.1 Photoionization cross sections

Both codes use Verner et al. (1996) fits with the Verner & Yakovlev (1995) inner-shell data as a fallback, and both take H I and He II from the hydrogenic analytic form. EXHALE adds two cross sections that MoCHII does not carry: the metastable He I 2³S level (bridged onto the VFKY96 wing) and molecular hydrogen (Yan, Sadeghpour & Dalgarno 1998). MoCHII covers a broader element list (including Ne, Cl, and Ar); EXHALE instead carries the alkali species (Na, K) relevant to planet atmospheres.

2.2 Recombination

Before 2026-07-17 the EXHALE H/He case-B coefficients were unattributed fits inherited from ATES. A numerical check showed them to be exactly the Hui & Gnedin (1997) case-B fits. MoCHII builds its default case-B coefficients from the Badnell (2023) radiative α_A and the Mao & Kaastra (2016) α_1 subtraction, with Badnell dielectronic recombination added for He II. The 2026-07-17 update (Part 2) adopts the same construction as the EXHALE default, so the two codes now agree on the H/He recombination coefficients. For metals, both codes already used Badnell radiative and dielectronic recombination, with small differences in the fallback data for a few ions.

2.3 Collisional ionization

Before the update the EXHALE H I and He I collisional-ionization rates came from Abel et al. (1997), which reproduces the Janev et al. (1987) fits, and He II from a Hui & Gnedin (1997) fit; none of these carried a source annotation. MoCHII uses the unified Voronov (1997) fits throughout, with Dere (2007) as an option. The 2026-07-17 update makes Voronov (1997) the EXHALE default for H and He as well; the metal rates were already Voronov (1997) in both codes.

2.4 Charge exchange

Both codes transcribe Table 4 of Huang et al. (2023, ApJ 951, 123), which rests on Kingdon & Ferland (1996). EXHALE carries 63 of its 65 rows (23 in Group A, 2 in B, 6 in C, 32 in D; the two missing ones are in the metal-metal set). Two switches select them. The 23 metal+H reactions of Group A are on by default and `cx_full` adds the metal+He (C) and metal+metal (D) reactions. The He $\leftrightarrow$ H pair (B1, B2) is separate: it is not in the generic reaction list at all but is applied by dedicated routines in every ionization system containing He, gated by its own input key `He_H_charge_exchange` (default True) and independent of `cx_full`. MoCHII carries the metal reactions only.

The printed Table 4 exchanges the reactant labels of its two oxygen rows: the $\exp(-227/T)$ Boltzmann factor appears on $O^+ + H^0$, but $IP(O\text{I}) > IP(H\text{I})$ makes $O^0 + H^+$ the endothermic direction that must carry it. EXHALE assigns the two

Table 1: Atomic data for the processes EXHALE and MoCHII have in common. Both codes descend from the same published sources for each process; the right two columns note where they diverge. EXHALE rows marked “[updated]” reflect the 2026-07-17 default (Part 2).

Process	EXHALE	MoCHII
Photoionization cross sections	Verner et al. 1996 (VFKY96) with Verner & Yakovlev 1995 fallback; H I and He II from the hydrogenic analytic form. Adds He I 2 ³ S (VFKY96 wing bridge) and H ₂ (Yan, Sadeghpour & Dalgarno 1998). Tuned for planet atmospheres (Na, K).	Verner et al. 1996 with Verner & Yakovlev 1995 fallback; H I and He II hydrogenic. Wider element set (Ne, Cl, Ar).
Recombination (H, He)	[updated] Badnell (2023) radiative α_A minus Mao & Kaastra (2016) α_1 , giving case B; He II includes Badnell dielectronic recombination. Identical coefficients to MoCHII.	Badnell (2023) radiative α_A and Mao & Kaastra (2016) α_1 for case B, with Badnell dielectronic recombination for He II.
Recombination (metals)	Badnell radiative/dielectronic recombination (Badnell 2006 and adf48 data), Ca I from Shull & Van Steenberg 1982, Fe from the Huang et al. 2023 fit (Badnell covers neither the K-like nor the Mn-like/Cr-like sequences).	Badnell radiative/dielectronic recombination (badnell_rr/dr.dat) with a CHIANTI fallback.
Collisional ionization (H, He)	[updated] Voronov 1997.	Voronov 1997 (unified fits), with Dere 2007 as an option.
Collisional ionization (metals)	Voronov 1997.	Voronov 1997.
Charge exchange	Huang et al. 2023 (ApJ 951, 123) Table 4, based on Kingdon & Ferland 1996. 63 reactions. On by default: the 23 metal+H reactions of Group A, and the He $\leftrightarrow$ H pair of Group B under its own key He_H_charge_exchange (default True, independent of cx_full). cx_full adds the metal+He and metal+metal reactions of Groups C–D.	Huang et al. 2023 Table 4 (Kingdon & Ferland 1996), metals only.
Recombination cooling	Hui & Gnedin 1997 case-B cooling fits.	Hui & Gnedin 1997 lineage, with a case-A/case-B choice.
Free-free cooling	[updated] Constant 1.426×10^{-27} ; thermally-averaged Gaunt factor from van Hoof et al. 2014 (161-point table).	Constant 1.426×10^{-27} ; Gaunt from Hummer 1988 / van Hoof et al. 2014.
Metal line cooling	CHIANTI v11.0.2 fit $\Lambda(T) = T^{-1/2} \sum_i A_i e^{-T_i/T}$ (cooling_data/fit_cno_formulas.py), extended with density-dependent treatments (see §2.7).	CHIANTI v11.0.2 fit of the same form (chianti_cooling.py), in the low-density limit; emission lines handled by a separate n-level solver.
Heating (photoelectric)	[updated] Shull & van Steenberg (1985) secondary ionization in the main loop: a photoelectron with $E_0 > 40$ eV deposits only $f_{\text{heat}}(x)$ of its excess as heat and drives H I / He I secondary ionizations (default on, use_sec_ion). Below 40 eV the photoelectron thermalizes fully.	Full photoelectron excess energy thermalized; no secondary ionization.

rate coefficients to the physically correct rows; see HUANG2023_TABLE4_OXYGEN_ERRATUM.md for the detailed-balance and Cloudy cross-checks. Any code transcribing the table as printed inherits the error.

2.5 Cooling

Recombination cooling in both codes follows the Hui & Gnedin (1997) case-B family (MoCHII offers a case-A/case-B choice). Both use the same free-free constant, 1.426×10^{-27} ; the Gaunt factor is discussed in §3.4. Metal line cooling in both codes is a CHIANTI v11.0.2 fit of the form $\Lambda(T) = T^{-1/2} \sum_i A_i e^{-T_i/T}$ (EXHALE: `cooling_data/fit_cno_formulas.py`; MoCHII: `chianti_cooling.py`). The MoCHII fit is the low-density limit, with the emission lines themselves computed by a separate n-level solver.

2.6 Heating

EXHALE now includes Shull & van Steenberg (1985) secondary ionization by fast photoelectrons in the main loop (default on, `use_sec_ion`; see Part 2). MoCHII thermalizes the full excess energy of the photoelectron and does not add secondary ionization, which is appropriate in the nearly fully ionized H II region it targets. The `wind_ae` initial-condition generator separately uses the same SvS85 partition together with the Dere (2007) coefficients.

2.7 Processes carried by only one code

EXHALE carries several networks that MoCHII does not: the metastable He I 2^3S chemistry (Oklopčić & Hirata 2018; Bray 2000; Penning ionization from Taylor et al. 2025), the H ($n=2$) $2s/2p$ populations (Christie, Arras & Li 2013; Draine 2011), Lyman- α radiative transfer (Neufeld 1990), and the H_2/H_3^+ molecular network (Miller et al. 2013). Its cooling also adds density-dependent treatments absent from the MoCHII low-density fits: a two-dimensional (T, n_e) statistical equilibrium table for Fe II, the ground terms of C I, C II, N II and O I solved in statistical equilibrium (the origin of the [C II] $158 \mu\text{m}$ and [O I] $63 \mu\text{m}$ saturation), the Mg II h&k, Ca II H&K, and Na I D two-level terms, and a Fe I table from NIST.

MoCHII in turn carries an n-level statistical-equilibrium emission-line solver (CHIANTI v11), Storey & Hummer (1995) recombination lines, Porter et al. (2012, 2013) He I case-B emissivities, the nebular continuum (free-bound plus two-photon, from Nussbaumer & Schmutz 1984 / Almog & Netzer 1989), and dust photoelectric heating (Bakes & Tielens 1994).

One quantity crosses between them. MoCHII transports the He I 2^1S two-photon continuum as sampled packets and so needs the full Drake, Victor & Dalgarno (1969) shape; EXHALE never forms a spectrum, but its He II cascade coupling (`he_rec_coupling`) needs two moments of that same shape above the H I edge — how many ionizing photons a 2^1S decay yields and how much energy each of them deposits. Both had been round numbers, 0.56 and 3.0 eV. Integrating the shape gives 0.5564 and 2.512 eV: the photon count was right, but the deposited energy was 20% high, being close to the 3.511 eV a uniform in-band distribution would give rather than to the peaked one the pair actually has. MoCHII found the same defect in its own sampler, which drew the in-band photons flat. EXHALE now carries the two integrals as `f_2q_HeI` and `Ee_2q_HeI`.

3. Part 2 — The 2026-07-17 rate update

The changes below are implemented in the radiation modules `Cool_coeff.f90` and `util_ion_eq.f90`, in the temperature solver `T_equation.f90`, and in the input parsing under `src/modules/files_IO/`. Items 1, 2, and 3 are gated by the new `legacy_hhe_rates` switch; items 4 and 5 apply always, independent of the switch.

3.1 H/He recombination default

The default H/He case-B coefficients are now

$$\alpha_B = \alpha_A^{\text{Badnell}} - \alpha_1^{\text{Mao}}, \quad (1)$$

where $\alpha_A^{\text{Badnell}}$ is the Badnell radiative total (with the three-term Badnell dielectronic recombination added for He II) and α_1^{Mao} is the Mao & Kaastra (2016, A&A 587, A84) ground-state coefficient. The implementation is in `Cool_coeff.f90`: `rr_badnell`, `dr_HeII_badnell`, `rr_mao`, and the assembled `alphaB_HII_new / alphaB_HeII_new / alphaB_HeIII_new`. The coefficients match those in the MoCHII `recomb_mod.f90`.

3.2 H/He collisional ionization default

The default H/He collisional-ionization rates are now the Voronov (1997, ADNDT 65, 1) fits, implemented through `voronov_ci` (`Cool_coeff.f90`). The adopted parameters are listed in Table 2.

Table 2: Voronov (1997) collisional-ionization parameters adopted as the EXHALE default.

Ion	ΔE [eV]	P	A	X	K
HI	13.6	0	2.91×10^{-8}	0.232	0.39
HeI	24.6	0	1.75×10^{-8}	0.180	0.35
HeII	54.4	1	2.05×10^{-9}	0.265	0.25

3.3 The `legacy_hhe_rates` switch

A new input flag `legacy_hhe_rates` (default `False`) restores the previous fits: Hui & Gnedin (1997) recombination and the Abel et al. (1997) / Hui & Gnedin (1997) collisional ionization. It is declared in `parameters.f90`, defaulted and parsed in `input_read.f90`, and echoed to `parse_dump.txt` by `write_setup_report.f90` — grep the flag name in each; the line numbers this paragraph used to quote had gone stale. It is registered as row K14b in `docs/input_schema.md`. Setting it to `True` reverts items 1 and 2 exactly; the free-free changes (items 4 and 5) are not affected.

3.4 Free-free Gaunt factor

The free-free Gaunt factor is now the thermally-averaged non-relativistic $\langle g_{\text{ff}} \rangle(\gamma^2)$ of van Hoof et al. (2014, MNRAS 444, 420), with $\gamma^2 = Z_{\text{ion}}^2 \text{Ry}/kT$. The frequency average

$$\langle g_{\text{ff}} \rangle(\gamma^2) = \frac{\int e^{-u} g_{\text{ff}}(\gamma^2, u) du}{\int e^{-u} du} \quad (2)$$

is tabulated at 161 points in $\log_{10} \gamma^2 = -6$ to 10 in steps of 0.1 dex (generated by `cooling_data/gauntff_thermal_avg.py`; the table is `gff_avg` in `Cool_coeff.f90`). This replaces the previous unattributed two-branch logarithmic fit and applies always, regardless of `legacy_hhe_rates`. The tabulated value at $T = 10^4$ K and $Z = 1$ is $\langle g_{\text{ff}} \rangle = 1.266$.

3.5 Free-free charge-weighting correction

The previous code weighted He II (net charge +1) by $Z^2 = 4$ and evaluated the metal Gaunt factor at the nuclear atomic number ($Z = 8$ for O, $Z = 26$ for Fe). This is physically wrong: free-free emission scales with the ion *net* charge Z_{ion} , not the nuclear number. The correction sets $Z_{\text{ion}} = 1$ for H II, He II, and singly-ionized metals, and $Z_{\text{ion}} = 2$ for He III and doubly-ionized metals (`util_ion_eq.f90 eval_cool`, lines 376–396; `T_equation.f90`, lines 105–121). The most visible consequence is that the free-free cooling of a He II-dominated layer drops by a factor of four. Like the Gaunt-table change, this correction applies always.

3.6 Benchmark values

Table 3 lists the new default case-B recombination and collisional-ionization rates at $T = 10^4, 10^5, 10^6$ K, with the legacy values at 10^4 K where relevant. The Fortran implementation reproduces an independent Python recomputation to all printed digits, and the legacy branch reproduces the pre-update values exactly.

Table 3: New default rates (α_B and collisional ionization ci ; units $\text{cm}^3 \text{s}^{-1}$), with legacy values at 10^4 K for comparison. The legacy $\alpha_B(\text{HII})$ at 10^4 K is the Hui & Gnedin (1997) fit, 2.592×10^{-13} .

Coefficient	$T = 10^4$ K	$T = 10^5$ K	$T = 10^6$ K	legacy (10^4 K)
$\alpha_B(\text{HII})$	2.606×10^{-13}	3.070×10^{-14}	2.163×10^{-15}	2.592×10^{-13}
$\alpha_B(\text{HeII})$	2.807×10^{-13}	5.006×10^{-13}	1.015×10^{-12}	2.616×10^{-13}
$\alpha_B(\text{HeIII})$	1.536×10^{-12}	2.370×10^{-13}	2.244×10^{-14}	—
$ci(\text{HI})$	7.448×10^{-16}	3.963×10^{-9}	3.103×10^{-8}	7.247×10^{-16}

The new $\alpha_B(\text{HII})$ at 10^4 K differs from the legacy Hui & Gnedin (1997) value by about 1%; the He II and He III case-B coefficients differ more, mainly because the new He II value carries the Badnell dielectronic contribution explicitly.

3.7 Secondary ionization in the main loop

A photoelectron ejected with kinetic energy $E_0 = e_\nu - E_{\text{th}}$ above about 40 eV does not thermalize immediately: before it does, it collisionally ionizes further H I and He I atoms. Shull & van Steenberg (1985) give asymptotic fits for the

partition of the photoelectron's energy as a function of the ionized fraction x of the H+He nuclei,

$$f_{\text{heat}}(x) = 0.9971 (1 - (1 - x^{0.2663})^{1.3163}), \quad (3)$$

$$f_{\text{ion,HI}}(x) = 0.3908 (1 - x^{0.4092})^{1.7592}, \quad (4)$$

$$f_{\text{ion,HeI}}(x) = 0.0554 (1 - x^{0.4614})^{1.6660}. \quad (5)$$

As $x \rightarrow 1$ the heating fraction approaches unity and both ionization fractions go to zero; as $x \rightarrow 0$ the heating fraction vanishes and the ionization fractions reach their maxima (0.3908 and 0.0554). In neutral gas a large share of the photoelectron energy therefore goes into secondary ionizations rather than heat.

A 40 eV threshold gates the partition: for $E_0 \leq 40$ eV the photoelectron thermalizes fully (no secondary ionization), matching the wind_ae X-ray cutoff. SvS85 is strictly an $E_0 \gtrsim 100$ eV asymptotic fit, so applying it down to 40 eV is a deliberate approximation adopted here.

Only HI and HeI receive secondary ionizations, the two channels SvS85 resolves. The SvS85 Ly α excitation channel is assumed to escape as line radiation and is not put into any rate (a future refinement could couple it to the Ly α field of the excited-H model). Each absorbing species s (HI, HeI, HeII, the HeI triplet when active, H₂ when present, and the photoionizable metal ions) contributes photoelectrons of energy $E_0 = e_\nu - E_{\text{th},s}$ that drive $f_{\text{ion,HI}} E_0 / E_{\text{th,HI}}$ secondary HI ionizations and $f_{\text{ion,HeI}} E_0 / E_{\text{th,HeI}}$ secondary HeI ionizations. The HeI triplet (2^3S , threshold 4.8 eV) previously contributed opacity and its photoionization rate but no photoheating; its photoelectron energy now enters the heating and absorbed-energy budgets and the secondary source like every other absorber.

This differs from how wind_ae handles the same physics. wind_ae combines the SvS85 $f_{\text{heat}} / f_{\text{exc}}$ partition with the Dere (2007) coefficient tables to split the secondary ionizations across species, but those Dere tables are tied to the fixed wind_ae spectral grid and cannot be reused in the main loop, which runs on a runtime SED and a variable energy grid. The main loop therefore applies the species-resolved SvS85 fits ($f_{\text{ion,HI}}$, $f_{\text{ion,HeI}}$) directly.

The feature is gated by use_sec_ion (default True). With use_sec_ion = False the photoelectron thermalizes fully and the heating and photoionization rates are bit-identical to the legacy path. It is implemented in the PH_heat_H and PH_heat_HHe routines of util_ion_eq.f90, wired through the callers in ionization_equilibrium.f90 and post_process_adv.f90; the flag is declared in parameters.f90, parsed in input_read.f90, echoed by write_setup_report.f90, and documented as key K14c in input_schema.md.

3.8 Effect on existing results

Because the default rates change, results computed with the new default differ from the pre-update golden and gate outputs. Setting legacy_hhe_rates = True reproduces the old H/He rates and recovers the previous ionization balance, with two exceptions that apply in all cases: the free-free charge-weighting correction (§3.5) and the van Hoof et al. (2014) Gaunt table (§3.4). Those two changes are physical corrections and are intentionally not tied to the switch. Secondary ionization (§3.7) is a further default change on top of the rate update; use_sec_ion = False restores the full-thermalization heating and photoionization rates bit-identically.